Video Transcript

Video 8: Applying Naïve Bayes Theorem to Detect Spam Email (01:43)

So, let's see how we apply Bayes rule to a real problem. Consider that we're receiving emails and we want to filter out the spam. So, we want to apply a label to the incoming email. So, we're going to decide based on some features of that email. So, those features are going to be the words contained in the email. So, we're trying to make a decision about the label that we're going to apply, and the two labels that we're going to go with are spam or normal, based on the features. Now, normally with Bayes, we'll have that we can flip this around now and say, given a certain label, how often does that feature occur?

So, what's the probability of that feature given the label times the probability of the label. Now, the probability of the feature, if we have two labels, we can divide one by the other and that probability of the feature will drop out. We'll be left with this equation here that I'm showing, where the probability of it being normal given these features divided by the probability that it's spam given these features is given by this right hand side. So, that's one insight in that we don't need to worry about this division here.

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